

ARTUR IASENOVETS

Applied Cryptography / FHE Research Engineer

Chongqing, China • archee.busy@gmail.com • [LinkedIn](#) • [ORCID](#) • [Portfolio](#)

Master's researcher in Network & Information Security focused on fully homomorphic encryption (FHE), private information retrieval (PIR), and privacy-preserving systems. Hands-on experience building BGV-based PIR in Hyperledger Fabric using Lattigo and CKKS-based encrypted aggregation for federated reinforcement learning using TenSEAL, PyTorch, and NumPy. Strong fit for applied crypto/FHE roles requiring prototype development, empirical evaluation, and research-to-system translation.

Target Areas: FHE systems, private information retrieval, encrypted aggregation, privacy-preserving ML infrastructure, secure analytics, blockchain privacy.

Technical Skills

Cryptography: Fully Homomorphic Encryption, BGV, CKKS, PIR, searchable encryption, OPRF/OT-based constructions, differential privacy, secure aggregation, RLWE/LWE-based security assumptions

FHE Libraries: Lattigo, TenSEAL, Microsoft SEAL, OpenFHE; familiar with TFHE and Concrete documentation

Programming: Python, Go, TypeScript, JavaScript, SQL

ML / Data: PyTorch, NumPy, federated learning, reinforcement learning, intrusion-detection workflows

Systems / Security: Hyperledger Fabric, Docker, GitHub Actions, S3/MinIO, Airflow, YARA, Chainsaw, MITRE ATT&CK

Languages: [English C1](#), [Chinese HSK3](#), Russian native

Selected FHE / Applied Cryptography Projects

BGV-Based Private Information Retrieval for Hyperledger Fabric

- Implemented a single-server HE-PIR prototype in Go using Lattigo, with packed CTI records, client-side encrypted selection vectors, and chaincode-side ciphertext-plaintext evaluation.
- Designed fixed-window slot packing, record-capacity checks, channel-specific parameterization, and client-side decoding for privacy-preserving reads from Fabric world state.
- Benchmarked end-to-end latency, peer-side evaluation time, network traffic, storage overhead, and parameter/capacity constraints; reported 113 ms end-to-end latency and ~2 MB peer traffic per query in the prototype.
- **Stack:** Go, Lattigo, Hyperledger Fabric, Docker, BGV, PIR, RLWE-based HE.

CKKS-Based Encrypted Aggregation for Federated Reinforcement Learning

- Implemented CKKS-encrypted model-update aggregation using TenSEAL for a federated reinforcement learning network-defense prototype.
- Used chunking to encode model-update vectors into encrypted CKKS payloads; clients train locally in plaintext while aggregation is performed over encrypted updates.
- Compared plaintext, differential privacy, trust-weighted, HE, and combined HE+DAPI execution paths; focused on confidentiality of model updates rather than full encrypted training/inference.
- **Stack:** Python, TenSEAL, PyTorch, NumPy, CKKS, federated learning, reinforcement learning.

Publications & Manuscripts

Bringing Private Reads to Hyperledger Fabric via Private Information Retrieval

Iasenovets A., Tang F., Zhu H., Wang P., Liu L. — *arXiv preprint (submitted to IEEE)*

BGV-based PIR for private world-state reads in Hyperledger Fabric.

Nov. 2025

[arXiv:2511.02656](#)

Adaptive Privacy-Aware Federated Reinforcement Learning with Trust-Weighted Secure Aggregation

Manuscript in preparation — second author

CKKS-based encrypted aggregation using TenSEAL with chunked model-update encoding for privacy-preserving federated reinforcement learning.

2026

EVMK-SSE: Efficient and Verifiable Multi-Keyword Symmetric Searchable Encryption

Zhu H., Tang F., Qin W., Shan J., Wang P., Iasenovets A. — *J. King Saud Univ. Comput. Inf. Sci.*

Multi-keyword searchable encryption using a lightweight OT-based relaxed OPRF.

Dec. 2025

[DOI](#)

Experience

Graduate Student Researcher Chongqing University of Posts and Telecommunications — Applied Cryptography Group	Chongqing, China Sept. 2024 – Present
– Designed and implemented BGV-based private information retrieval for Hyperledger Fabric using Lattigo, including packed plaintext encoding, encrypted query generation, ciphertext-plaintext evaluation, and client-side decoding. – Implemented CKKS-based encrypted aggregation for federated reinforcement learning using TenSEAL, PyTorch, and NumPy, including chunked model-update encoding. – Evaluated privacy-preserving prototypes using latency, communication volume, storage overhead, ciphertext expansion, and parameter/capacity constraints. – Contributed to research papers on FHE-based PIR, searchable encryption, cyber threat intelligence, and privacy-preserving federated learning.	
Security Intern Positive Technologies — Leading Eastern European cybersecurity company	Remote Aug. 2024 – Feb. 2025
– Developed Python utilities for MaxPatrol EDR, including regex-based Windows path/network configuration parsing, unit-test coverage, and a Windows Prefetch-based allowlist/blocklist checker. – Built a malware-processing pipeline for Threat Intelligence: VX-Underground sample ingestion, S3/MinIO artifact storage, YARA scanning, and Airflow-based daily automation. – Conducted SOC analysis of CVE-2020-5902 and CVE-2023-38831 attack chains using Chainsaw, PowerShell deobfuscation, oletools, IoC extraction, and MITRE ATT&CK mapping.	
Software Engineer academai.ru — AI-powered EdTech startup under Sberbank's Accelerator program	Remote Aug. 2023 – Aug. 2024
– Built Flask REST API endpoints orchestrating structured-prompt LLM calls via Langchain to generate personalized course components. – Developed frontend functionality with Next.js, TypeScript, Prisma, and Auth.js; built Dockerized CI/CD pipelines with GitHub Actions.	

Education

Chongqing University of Posts and Telecommunications M.S. Network & Information Security Supervisor: Prof. Tang Fei GPA: 3.9/4.0	Chongqing, China Sept. 2024 – Present
Ukhta State Technical University B.S. Information Systems and Technologies GPA: 4.5/5.0 (3.6/4.0)	Russia March 2022 – June 2024
St. Petersburg State University of Telecommunications named after Prof. M.A. Bonch-Bruевич B.S. Information Systems and Technologies	Russia Sept. 2020 – March 2022

Selected Honors

Chinese Government Scholarship for Graduate Studies	Sept. 2024 – Aug. 2027
5 Science Conferences — 3 first-place paper evaluations at USTU	Apr. 2022 – Apr. 2023
Kaspersky Cyberimmune Development Hackathon 2.0 — 11th of 100+ teams	Dec. 2023
MIPT & Sber Phystech GigaChat Challenge Hackathon — LLM course-generation app	Dec. 2023